

HEINOLA ASEMANTAUS, Finland

WMO: 027680

Lat: 61.200N

Lon: 26.049E

Elev: 92

StdP: 100.22

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-20.7	-26.7	0.3	-23.8	-23.0	0.5	-20.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.1	18.2	26.1	17.3	24.2	16.5	20.2	25.1	18.9	23.8	17.7	22.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	22.1	17.2	12.4	20.9	16.0	11.5	19.4	58.4	25.5	54.2	23.9	50.4	22.2	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-24.7	29.5	4.8	2.5	-28.2	31.3	-31.0	32.8	-33.7	34.2	-37.2	36.0
			-24.9	20.8	4.8	1.8	-28.3	22.1	-31.1	23.2	-33.8	24.2	-37.3	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-7.6	-5.2	-2.1	4.0	10.9	14.8	17.8	16.0	11.2	4.8	1.3	-2.7
	DBStd	9.47	6.78	6.14	4.57	3.59	4.31	3.54	3.48	3.01	3.28	3.49	3.75	5.40
	HDD10.0	2422	545	427	376	184	41	4	0	1	26	164	261	394
	HDD18.3	4811	803	660	634	431	231	117	53	84	215	419	511	653
	CDD10.0	714	0	0	0	3	69	147	243	187	62	3	0	0
	CDD18.3	61	0	0	0	0	2	10	37	12	0	0	0	0
	CDH23.3	593	0	0	0	0	47	116	331	99	0	0	0	0
	CDH26.7	128	0	0	0	0	5	18	83	22	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	609	44	36	32	31	44	64	70	67	53	61	56	48
	PrecMax	787	86	81	59	60	110	100	136	119	131	149	131	95
	PrecMin	504	8	2	2	9	9	17	19	10	20	10	5	11
	PrecStd	74	19	21	15	14	25	24	35	30	27	32	26	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.4	10.0	19.9	27.1	28.8	31.2	29.4	21.6	13.9	10.4	8.4
		MCWB	2.8	2.6	4.4	10.4	14.6	17.5	19.8	19.0	16.0	11.4	9.1	7.6
	2%	DB	2.6	3.4	7.1	16.2	24.5	26.5	28.9	25.7	19.3	12.1	8.8	5.2
		MCWB	2.1	2.1	3.0	8.0	14.3	16.6	19.1	17.6	14.8	10.4	8.1	4.4
	5%	DB	1.6	2.2	5.0	13.3	21.9	24.2	27.0	23.7	17.6	11.2	7.2	4.0
		MCWB	1.2	1.5	2.3	6.9	13.2	15.5	18.6	16.5	13.8	9.8	6.6	3.3
	10%	DB	0.5	1.3	3.4	10.7	19.4	22.0	24.8	21.7	16.2	10.0	6.1	2.7
		MCWB	0.2	0.8	1.5	5.7	11.9	14.7	17.7	15.9	13.5	8.8	5.5	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.1	5.4	11.4	17.4	19.7	21.9	21.1	16.5	12.3	9.2	7.6
		MCDB	3.5	3.9	7.9	17.2	22.5	25.2	27.1	26.5	19.7	13.4	10.1	8.3
	2%	WB	2.2	2.3	3.8	9.5	15.7	18.0	20.7	19.2	15.6	11.1	8.0	4.7
		MCDB	2.6	3.1	6.3	15.2	21.0	23.8	26.0	23.1	18.0	11.8	8.7	5.2
	5%	WB	1.2	1.6	2.8	7.6	14.3	16.7	19.7	17.8	14.9	10.1	6.7	3.4
		MCDB	1.5	2.3	4.5	11.7	19.5	22.3	24.6	21.6	17.0	10.9	7.1	3.9
	10%	WB	0.3	0.9	1.8	6.3	12.9	15.5	18.7	16.9	13.9	8.8	5.7	2.2
		MCDB	0.6	1.3	3.2	9.9	19.4	20.6	23.5	20.4	15.6	9.7	6.0	2.5
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.3	7.4	9.2	10.7	9.5	9.3	8.7	7.1	5.2	3.5	3.8
		MCDBR	3.4	4.6	8.7	14.2	15.2	13.6	12.5	12.2	9.4	6.5	4.4	4.6
		MCWBR	3.3	3.8	5.6	7.5	6.5	5.3	4.3	5.1	5.5	4.8	4.2	4.4
	5% WB	MCDBR	3.6	4.3	7.5	11.6	12.6	11.5	10.9	10.0	7.9	5.7	4.3	4.3
		MCWBR	3.6	3.8	5.2	6.9	6.2	5.2	4.6	4.9	5.1	4.9	4.1	4.2
Clear-Sky Solar Irradiance	taub	0.255	0.297	0.324	0.364	0.361	0.356	0.379	0.366	0.342	0.322	0.279	0.366	
	taud	2.087	2.156	2.234	2.340	2.417	2.457	2.409	2.437	2.497	2.408	2.179	2.603	
	Ebn at Noon	456	666	787	828	862	872	838	820	778	652	425	306	
	Edh at Noon	39	71	92	102	102	100	102	92	73	56	36	25	
All-Sky Solar Radiation	RadAvg	0.23	0.81	2.29	3.74	5.06	5.30	5.18	3.99	2.34	0.98	0.27	0.12	
	RadStd	0.04	0.18	0.29	0.40	0.56	0.37	0.51	0.46	0.29	0.17	0.05	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	-155	-170	N/A	N/A	

Nomenclature: See separate page